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Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

RFP-TELTECH-FWA-2026-001 / APPENDIX C

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Service Level Agreements (SLAs), Network KPIs & Liquidated Damages

Operational Performance Benchmarks, Severity MTTR & Financial Penalty Schedules

DOCUMENT NUMBER	AUTHOR	TARGET SPECTRUM	BIDDING STATUS
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1. OBJECTIVE & GOVERNANCE PRINCIPLES

This Appendix defines the mandatory Service Level Agreements (SLAs), Key Performance Indicators (KPIs), measurement methodologies, reporting intervals, and financial remedies (Service Credits and Liquidated Damages) governing:

1. **Turnkey Deployment & Commissioning Phases (Lots 1, 2, 5)**
2. **5G Radio & Core Network Operational Performance (Lots 1, 3)**
3. **OSS/BSS Platform Reliability & Field Dispatch Operations (Lot 4)**
4. **Managed Services & First Line Maintenance Operations (Lot 6)**

Performance shall be evaluated on a calendar monthly basis using the automated performance management OSS and audited ticketing records.

2. PROJECT ROLLOUT & DELIVERY SLAS

2.1 Milestone Delivery Schedule

The Contractor shall adhere strictly to the following turnkey delivery schedule for Phase 1 (1,000 sites):

* **Milestone 1 (TSSR & Nominal Engineering Approval):** Within **45 Calendar Days** of Notice to Proceed (NTP — January 15, 2027) for all 1,000 priority sites.
* **Milestone 2 (First 200 Sites Commercial On-Air):** Within **120 Calendar Days** of Notice to Proceed (NTP).
* **Milestone 3 (Cumulative 500 Sites Commercial On-Air):** Within **210 Calendar Days** of Notice to Proceed (NTP).
* **Milestone 4 (Cumulative 1,000 Sites Commercial On-Air & Integrated):** Within **365 Calendar Days (12 Months)** of Notice to Proceed (NTP).

2.2 Delay Liquidated Damages (Rollout)

- If the Contractor fails to achieve Final Acceptance of a milestone by the scheduled completion date, the Operator shall deduct **Delay Liquidated Damages (DLD)** equal to **0.5% of the total milestone contract value per calendar week of delay**, up to a maximum aggregate cap of **10.0% of the total contract value**.
- Delays exceeding 60 calendar days beyond the milestone date shall entitle the Operator to terminate for default and draw down on the Contractor's Performance Bond.

3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIS & SERVICE CREDITS

3.1 Core Radio KPIs (Calculated 24/7/365 Nationwide)

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Radio Availability (%) = [(Total Operating Cell Hours – Total Unplanned Outage Cell Hours) / Total Operating Cell Hours] * 100
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KPI IDENTIFIER	METRIC DESCRIPTION	MEASUREMENT METHODOLOGY	TARGET STANDARD	MINIMUM THRESHOLD	SERVICE CREDIT PENALTY (< THRESHOLD)
KPI-RAN-01	**Radio Network Availability**	Monthly aggregate across all active cells	**>= 99.95%**	**99.70%**	1.0% of monthly O&M fee per 0.1% shortfall
KPI-RAN-02	**5G Session Setup Success (CSSR)**	Initial RRC connection + 5G QoS flow setup (RRC x 5G QoS Flow)	**>= 97.00%**	**96.00% (Min)**	0.5% of monthly O&M fee per 0.2% shortfall below 96.00%
KPI-RAN-03	**5G Session Drop Rate (CDDR)**	Abnormal session drops during active data sessions	**<= 1.50%**	**2.00% (Max)**	0.5% of monthly O&M fee per 0.1% excess above 2.00%
KPI-RAN-04	**Loaded Cluster Sector DL Throughput**	Busy hour average sector throughput across cluster	**>= 450 Mbps**	**380 Mbps**	Mandatory RF re-tuning within 5 days
KPI-RAN-05	**Cell-Edge DL User Speed**	5th percentile user speed (>= 95% of test points)	**>= 50 Mbps**	**25 Mbps**	Mandatory antenna tilt optimization; SSV governed
KPI-RAN-06	**Cell-Edge UL User Speed**	5th percentile user speed (>= 95% of test points)	**>= 15 Mbps**	**5 Mbps**	Uplink power control re-configuration; SSV governed
KPI-RAN-07	**Radio Round-Trip Latency (RTT)**	CPE to local gNodeB/UPF edge	**<= 18 ms**	**30 ms**	Audit packet buffering & backhaul
KPI-RAN-08	**Packet Loss Rate (PLR)**	PDCP Layer packet discard rate per 3GPP TS 38.314	**<= 0.10%**	**0.20% (Max)**	Transport & radio QoS re-profiling; Service Credits if > 0.20%

3.2 Service Credit Assessment & Stabilization Governance (Bid Bulletin No. 2 Harmonization)

1. **Financial Penalty Floor:** Service Credit liquidated damages are assessed **exclusively when monthly performance breaches the Minimum Threshold floor** (or exceeds the Maximum Threshold for CDDR/PLR). Performance falling between the Target Standard and Minimum/Maximum Threshold mandates engineering root-cause analysis and corrective cluster optimization without financial penalty.
2. **Post-PAC Cluster Stabilization Window:** A **90-day stabilization grace period** is granted for each site rollout cluster commencing from Provisional Acceptance (PAC), during which financial Service Credits for Network Availability (NAV) and MTTR are suspended while operational baselines and field spares logistics are stabilized.
3. **OSS Throughput Counters vs Contractual SLAs:** Instantaneous OSS cell-edge throughput counters shall not incur financial penalties due to dependency on live subscriber distribution and concurrency. Cell-edge speed compliance is governed by Single Site Verification (SSV) and cluster drive acceptance.

4. 5G CORE NETWORK & CLOUD PLATFORM KPIS

4.1 5GC High Availability Benchmarks

KPI IDENTIFIER	METRIC DESCRIPTION	MEASUREMENT TARGET	CALCULATION INTERVAL	LIQUIDATED REMEDY
KPI-5GC-01	Control Plane NF Availability (AMF/SMF/UDM)	>= 99.999% ("Five Nines")	Monthly	$10,000 USD per hour outage exceeding threshold
KPI-5GC-02	User Plane (UPF) Packet Availability	>= 99.999%	Monthly	$10,000 USD per hour outage exceeding threshold
KPI-5GC-03	PDU Session Establishment Success Rate	>= 99.80%	24-hr Rolling	Root-cause analysis within 12 hours
KPI-5GC-04	CG-NAT Port Allocation Success Rate	>= 99.99%	Monthly	$5,000 USD penalty per incident causing unallocated subscriber sessions exceeding 10 minutes
KPI-5GC-05	Session-Preserving Active-Active Failover Time	<= 30 Seconds (Zero session state loss; user traffic restored <= 30s)	Semi-annual Drill	Penalty of 5% annual core AMC if drill fails
KPI-5GC-06	UPF Internal Packet Latency	<= 1.0 ms	Continuous	Hardware NIC DPDK tuning

5. OSS/BSS & FIELD OPERATIONS SLAS

5.1 Commercial & Operational Velocity Metrics

KPI IDENTIFIER	SERVICE LEVEL DOMAIN	MANDATORY TARGET	IMPACT ON FAILURE
KPI-BSS-01	Converged Charging System (CHF) Uptime	>= 99.99%	2.0% monthly BSS maintenance fee deduction
KPI-BSS-02	Top-Up Processing Speed (GCash / Maya)	<= 3.0 Seconds	Mandatory optimization of API gateway queues
KPI-OSS-01	New Installation Dispatch Lead Time	<= 48 Hours	>= 95% of orders completed within 48h SLA
KPI-OSS-02	First-Time-Right (FTR) Installation Rate	>= 95.0%	Repeat truck rolls at Contractor expense
KPI-OSS-03	Churned CPE Hardware Recovery Rate	>= 70.0%	Contractor debited $55 USD per unrecovered unit below 70%
KPI-OSS-04	Automated Work-Order Dispatch Engine Capacity	>= 3,500orders/day	Zero backlog accumulation during peak campaigns

6. MANAGED SERVICES & FIELD INCIDENT RESTORATION SLAS

6.1 Incident Classification & Mean Time to Restore (MTTR)

Incidents logged by the 24/7 NOC are classified into four severity tiers:

SEVERITY LEVEL	DEFINITION & OPERATIONAL IMPACT	RESPONSE SLA	MTTR SLA (URBAN)	MTTR SLA (RURAL)
Severity 1 (Critical)	Complete macro site outage; Core NF failure; >500 subscribers affected; major backhaul failure.	<= 15 Mins	<= 2.0 Hours	<= 4.0 Hours

SEVERITY LEVEL	DEFINITION & OPERATIONAL IMPACT	RESPONSE SLA	MTTR SLA (URBAN)	MTTR SLA (RURAL)
Severity 2 (Major)	Single sector down; degraded throughput; loss of DC battery backup; generator fault.	<= 30 Mins	<= 4.0 Hours	<= 6.0 Hours
Severity 3 (Minor)	Non-service affecting hardware fault; loss of redundancy (single PSU failed, SFP degraded).	<= 1 Hour	<= 12.0 Hours	<= 24.0 Hours
Severity 4 (Warning/Info)	Configuration inquiry, routine audit, planned preventative maintenance scheduling.	<= 4 Hours	<= 48.0 Hours	<= 72.0 Hours

6.2 Service Credits for MTTR Breaches

For every hour (or fraction thereof) that a Severity 1 incident exceeds the mandated MTTR threshold, the Operator shall deduct:

* **Urban / Suburban Sites: 0.5% of monthly site Managed Services fee per hour** of prolonged outage.
* **Rural Sites: 0.25% of monthly site Managed Services fee per hour** of prolonged outage.
* Total monthly service credit deductions for Managed Services shall be capped at **20.0% of the total monthly Managed Services fee**.

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7. MONTHLY REPORTING & GOVERNANCE CADENCE

1. **Daily Operational Flash Report:** Generated automatically at 06:00 PHT summarizing the previous 24 hours of network availability, outages, MTTR compliance, and work orders.
2. **Weekly Engineering Review:** Joint operations meeting reviewing open trouble tickets, cluster drive tests, and rollout milestone progress.
3. **Monthly Governance & SLA True-Up:** Formal executive session reviewing all KPI metrics against SLA thresholds, finalizing service credits, and approving contractor invoices.

(End of Appendix C — Service Level Agreements, Network KPIs & Liquidated Damages)
